

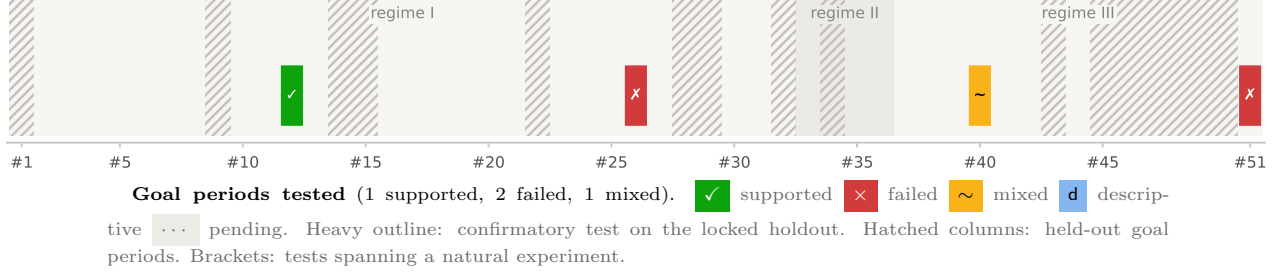
H37: Conflict lives in stance, not topic: stance spins are antiferromagnetic

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2.5/5

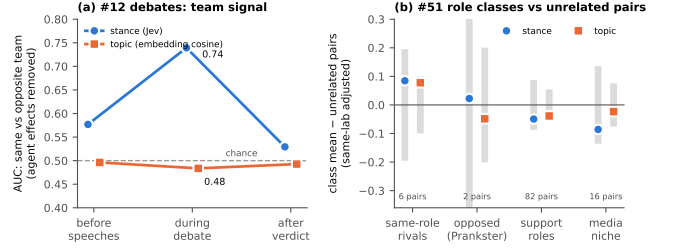
Bottom line. A validated disagreement flag sees the conflict a debate assigns (45% vs 0% of replies; topic blind), and rival and opposed agents show none: disagreement lives only where imposed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H22 finds that rival agents in #51 co-move in *topic*, so topic cannot show conflict. H37 tests whether *stance* shows it. A zero-shot labeller (Jev) marks 30k reply pairs in #12, #26, #40 and #51. In the #12 debates, stance separates opponents from teammates (AUC 0.74 after agent effects), and topic does not (0.48). Stance recovers the drafted teams exactly in 7 of 10 debates, and the contrast vanishes after the verdict. #51 rivals and opposed roles are as friendly as anyone, and #26 stance ignores ballots. **Round 1c** re-tests this on ledger-visible DQ2 replies with DQ10's validated disagreement flag (precision 0.61–0.67, recall $\approx$ 0.6). with its noise in every null. #12 opponents disagree in 33 of 74 replies and teammates in 0 of 60 (stress null $p = 0.001$). #40 now raises no false alarm. #51 rivals and opposed roles show no excess disagreement. The #34 saboteur test is **not run**; a re-freeze on held-out v2.1 labels is proposed.



Observables. (a) #12: AUC for same-team against opposite-team replies, with agent effects removed. Stance (blue) separates the teams only while they debate. Topic similarity of the same reply pairs (orange) never does. (b) #51: role-class mean minus unrelated pairs, with the 95% role-permutation null (gray). Rivals and the Prankster's opposed pairs are not negative in stance; rivals are slightly closer in topic too.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To find assigned sides or a scapegoated agent, run a stance labeller on replies. It costs about \$1.12 per 30k reply pairs and recovers debate teams exactly in 7/10 debates. Check every camp alarm against the agent-field (ordered-logit) null, which holds 4% false alarms; the sign-shuffle null fires falsely in 28–100%. LLM agents almost never take sides unless a protocol tells them to, so the detector mostly confirms that nothing happens. Whether it catches covert sabotage is the open #34 question. Do not monitor the raw “negative reply” rate: it is mostly correction friction. Use the validated disagreement flag, corrected for precision $\approx$ 0.64 and recall $\approx$ 0.6.

PHYSICAL MODEL

Models 01 (signed Ising) and 10 (Potts communities). The spin is a bond: the stance $s_e \in [-1, 1]$ of reply e from j to i , from Jev probabilities. a_j and b_i are agent fields (agreeableness, likability) of an ordered logit with cut points c_k . Δ is the camp coupling. J_{ij} is the residual pair stance after the fields, and $\xi_i = \pm 1$ is the camp. Factions are the ground state of $H(\xi)$; frustration f is the weight it leaves unsatisfied. γ is same-team ($\delta_{ij}^{\text{team}} = 1$) minus opposite-team stance, with fields absorbed.

$$P(\sigma_e \leq k) = \text{logistic}(c_k - a_j - b_i - \Delta \xi_i \xi_j), \quad \sigma_e \in \{-, 0, +\}$$

$$s_e = p_{\text{agree}} + p_{\text{support}} - p_{\text{oppose}} - p_{\text{undermine}}$$

$$s_e = \mu + a_j + b_i + \gamma \delta_{ij}^{\text{team}} + \varepsilon_e$$

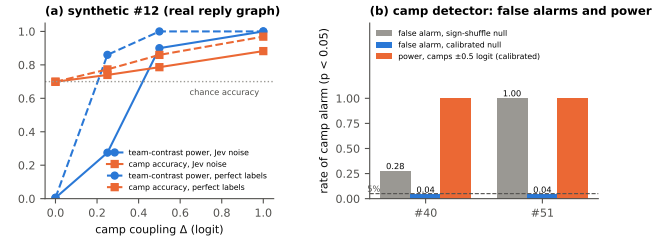
$$H(\xi) = - \sum_{i < j} J_{ij} \xi_i \xi_j$$

$$f = \frac{1}{2} \left(1 - \max_{\xi} \left[-H(\xi) \right] / \sum_{i < j} |J_{ij}| \right)$$

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 fra/summaries/build_summaries.py · card hypotheses/H37-stance-
 spins/README.md · ratings: H37 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1c (validated Verdict flag)
P1 #12 teams	γ 0.42 → flags 33/74 across vs 0/60 held within; stress p 0.001
P2 not topic	AUC 0.74 vs 0.48 → topic AUC 0.48; held holds with topic covariate
P5 verdict	γ 0.42→0.03 → 0 flags in 90 replies held
P6 #51 op-posed	T_{OP} +0.02 → 0/55 replies, $\beta = -2.0$ failed
P7 #51 rivals	T_{SR} +0.085 → $\beta = +0.46$ [-1.5, 1.4] failed
P10 #26 bal-lots	r 0.01 → 7 flags, power 0.05 untestable
P12 #40	20% “oppose” → true rate 0.05× de-mixed→held bate; 0 pairs



Second figure. (a) Synthetic #12 on the real reply graph. With Jev’s measured noise, the team contrast is detectable from $\Delta \approx 0.5$ logit; real #12 matches $\Delta \gtrsim 1$. (b) Camp alarms under agent fields only. The sign-shuffle null fires falsely (28% at #40, 100% at #51); the calibrated ordered-logit null holds 4%.

GOAL PERIODS IN DETAIL

G12 ✓ supported (positive control passes P1–P5)
 G26 × failed (stance does not track ballots; no factions)
 G40 ~ mixed (no false faction or pair alarm; conflict level...
 G51 × failed (no stance antagonism between rival or op-posed...)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Round 1 used one labeller and one rater, and Jev’s “oppose” was only 30% precise. Round 1c’s flag is validated (precision 0.61–0.67) but rare (0.85% in #51). Rival tests are therefore inconclusive at +1 log-odds. #51 rests on 6 rival and 2 opposed pairs; #26 has 9 voters. The calibrated null (Amendment 2) came after the data. The topic baseline uses one embedding model.

ROUND 2

Add a second labeller and a human rater. Run the calibrated detector on contested periods (#19, #31) and, pre-registered, on the #51 tail. Test signed reply dynamics: measure whether a negative reply begets another. Run the #34 saboteur confirmation after sign-off.

- **Essence.** a cheap, calibrated sensor for adversarial or factional behaviour in agent swarms, and its detection limits.
- **H37-R1** A second labeller and rater.
- **H37-R2** Find conflict where it might exist.
- **H37-R3** Kinetic stance.
- **H37-R4** Run the #34 confirmation after sign-off, as the saboteur test (D8.3).

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